

Project Presentation and Submission

▶ Project presentation

- ▶ Time: June.4,1-2, 203
- ▶ June.4,11-12, 203
- ▶ Each group has <7 minutes for presentation + 2 minutes QA
- ▶ Schedule:
 - ▶ 1-2: group 1-10
 - ▶ 11-12: group 11-21

▶ Report & Code

- ▶ Due: 11:59pm, June. 13
- ▶ Submission link at BB -> Project -> Project Material Submission
- ▶ Format: PDF, academic paper, no page requirement (Report); Code
- ▶ Each group only needs to submit one report and code



Final Exam

- ▶ Time
 - ▶ 8:00-10:00am, June. 9
- ▶ Location
 - ▶ 教学中心302
- ▶ Format
 - ▶ Closed-book. You can bring **an A4-size cheat sheet**.
 - ▶ 10 multiple-choices (不定项选择题), 4 problems
- ▶ Grade
 - ▶ 25% of the total grade





Final Review



Disclaimer

- ▶ Topics covered in this review may not appear in the exam.
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Bayesian networks: Inference

- ▶ Exact inference
 - ▶ Inference by enumeration
 - ▶ Variable elimination
 - ▶ Interleave join (pointwise product) and elimination (summing out)
 - ▶ Efficient inference on polytrees
- ▶ Approximation inference
 - ▶ Prior Sampling
 - ▶ Rejection Sampling
 - ▶ Likelihood Weighting
 - ▶ Gibbs Sampling



Probabilistic temporal models

- ▶ Markov models
 - ▶ Markov assumption, Transition model
- ▶ Hidden Markov models
 - ▶ Transition model (states) + emission model (evidence)
 - ▶ Filtering: $P(X_t | e_{1:t})$
 - ▶ Forward algorithm
 - ▶ Most likely explanation: $\operatorname{argmax}_{x_{1:t}} P(x_{1:t} | e_{1:t})$
 - ▶ Viterbi algorithm
- ▶ Dynamic Bayes networks
- ▶ Approximate inference by particle filtering
 - ▶ Propagate forward → Weight → Resample



Markov Decision Processes

- ▶ Markov Decision Process
 - ▶ States S , Actions A , Transitions $P(s'|s,a)$, Rewards $R(s,a,s')$
- ▶ Quantities:
 - ▶ Policy, Utility, Values, Q-Values
- ▶ Solve MDP
 - ▶ Bellman equation
 - ▶ Value iteration
 - ▶ Policy iteration
 - ▶ Policy evaluation + Policy improvement



Reinforcement Learning

- ▶ Reinforcement learning
 - ▶ MDP without knowing T and R
 - ▶ Offline planning vs. online learning
- ▶ Model-based learning
- ▶ Model-free learning
 - ▶ Policy evaluation: Temporal Difference Learning
 - ▶ Exponential moving average
 - ▶ Computing q -values/policy: Q-Learning
- ▶ Exploration vs. Exploitation
 - ▶ Random exploration, exploration function
- ▶ Approximate Q-Learning
 - ▶ Feature-based representation of states



Supervised machine learning

- ▶ To learn an unknown target function f from labeled examples
- ▶ Classification (f with discrete output value)
 - ▶ Naïve Bayes
 - ▶ All attributes are independent given class
 - ▶ Generalization and overfitting, smoothing
 - ▶ Perceptron (linear classifier), neural networks
- ▶ Regression (f with continuous output value)
 - ▶ Linear regression, minimizing summed squared error



Unsupervised machine learning

- ▶ K-means
 - ▶ Clustering
 - ▶ Iteration:
 - ▶ Assign each data instance to closest center
 - ▶ Assign each center to the average of its assigned data points
- ▶ Expectation-Maximization
 - ▶ Learning Mixture of Gaussians
 - ▶ Iteration:
 - ▶ E-step: Compute label distribution of each data point
 - ▶ M-step: Update each Gaussian based on its (proportionately) assigned points
- ▶ Spectral Clustering
 - ▶ Graph Cut / Normalization
 - ▶ Dimensionality Reduction



Transformer – Pretraining-Finetuning

- ▶ Transformer
 - ▶ Self-attention
 - ▶ Multi-head Attention
 - ▶ Transformer Layer
 - ▶ Transformer Encoder & Decoder
- ▶ Pre-training-Fine-tuning
 - ▶ Pre-training
 - ▶ Fine-tuning
 - ▶ ELMo, GPT, BERT





Good luck in your final exam 😊



The Road Forward

- ▶ Related Courses
 - ▶ Undergraduate
 - ▶ SI151 Optimization and machine learning
 - ▶ CS150 Database and Data Mining
 - ▶ CS172 Computer vision I
 - ▶ Graduate
 - ▶ CS280 Deep Learning
 - ▶ CS282 Machine Learning
 - ▶ CS243 Introduction to Algorithmic Game Theory
 - ▶ CS272 Computer Vision II
 - ▶ CS283 Robotics
 - ▶ CS284 Simultaneous Localization and Mapping
 - ▶ SI232 Subspace Learning
 - ▶ SI252 Reinforcement Learning
 - ▶ More to come...

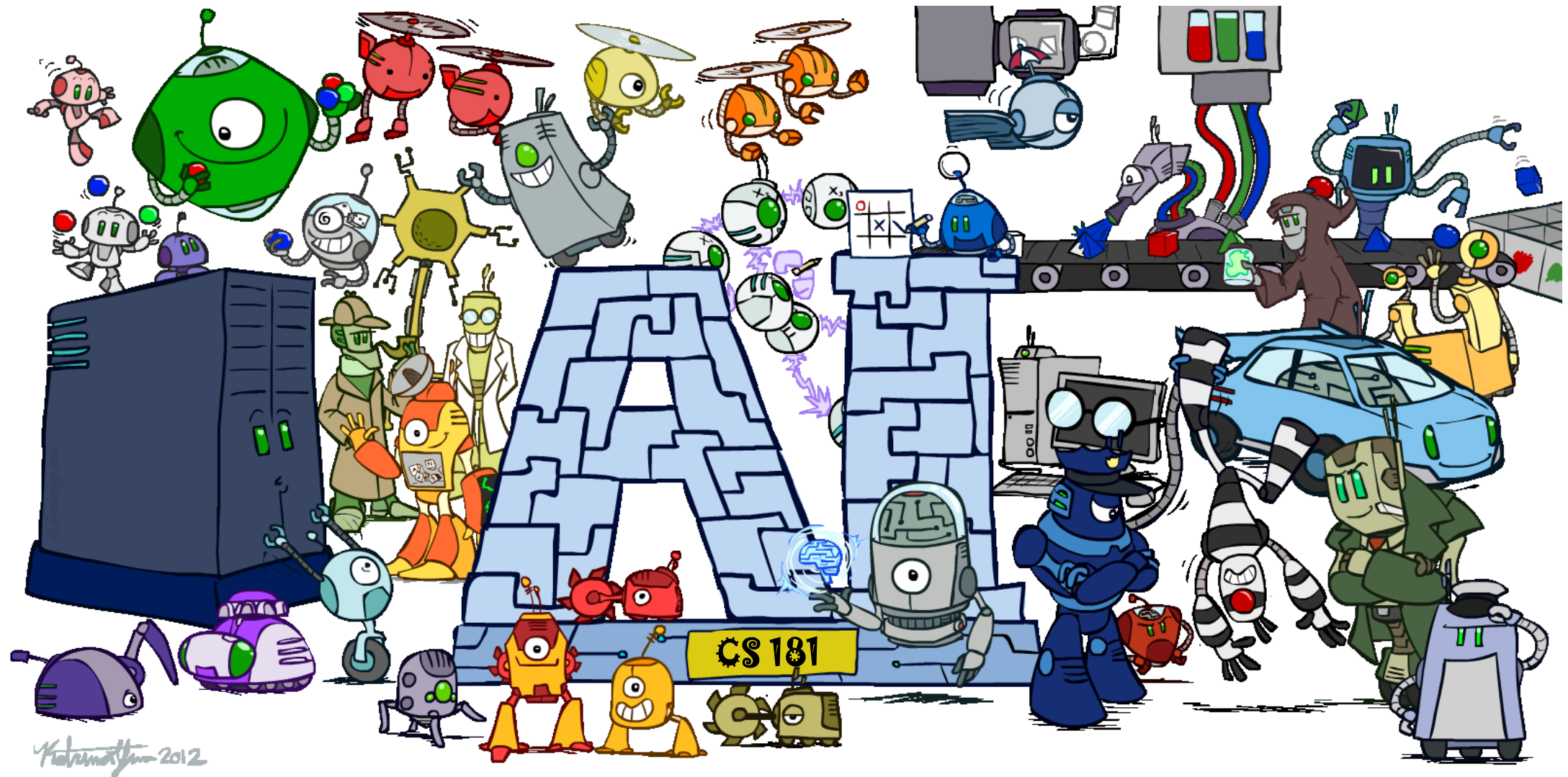


The Road Forward

- ▶ Learning recent developments in AI from top conferences
 - ▶ ML: NIPS, ICML, ICLR
 - ▶ CV: CVPR, ICCV, ECCV
 - ▶ NLP: ACL, EMNLP, NAACL
 - ▶ AI: IJCAI, AAAI
 - ▶ Caution: not top in ML, NLP, CV
 - ▶ Other: KDD, AAMAS, UAI, SIGIR, ...



That's all!



Krishna 2012